

Yumeng Cai

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EDUCATION

- Georgia Institute of Technology** (*M.S. in Robotics*) GPA: 3.82/4.0 Atlanta, GA, USA
Aug. 2025 - May 2027 (Expected)
- South China University of Technology** (*B.E. in Robotics Engineering*) GPA: 3.78/4.0 Guangzhou, China
Sep. 2021 - Jun. 2025
- Scholarship for Excellence (Second Class, 2022–2023; Third Class, 2021–2022 and 2023–2024)

PUBLICATIONS & PATENTS

- Yumeng Cai**[†], L. Gan[†], Y. Zhang, X. Wang, Y. Zhou, “A Miniature Hybrid-Actuated Soft Manipulator with FBG-Based Shape Sensing for Confined Medical Procedures,” submitted to *IEEE International Conference on Robotics and Biomimetics*, 2026.
- Y. Wang[†], **Yumeng Cai**[†], M. Pang, Y. Xu, G. Kassab, C. Zhou, G. Chase, X. Chen, H. Luo and Y. Chen, “Fluid-Structure Interaction of Left Ventricle with Impaired Ventricular Contractility,” *Acta Mechanica Sinica*, 2025.
- B. Xie, Y. Wang, **Yumeng Cai**, Z. Ouyang, C. Zhu, H. Sun, and Y. Chen, “Computational Fluid Dynamics Analysis of Tandem Floating Offshore Wind Turbines under Coupled Pitch-Surge Motion: A Comparative Study of the National Renewable Energy Laboratory 5 Megawatt and the International Energy Agency 22 Megawatt Turbines,” *Physics of Fluids*, 2025.
- Y. Wang[†], **Yumeng Cai**[†], B. Xie, C. Zhu, H. Luo, Y. Li, and Y. Chen, “A Computational Study of the Hydrodynamics of Quadrupedal Paddling,” *Biomimetics*, 2024.
- L. Gan, Y. Xu, **Yumeng Cai**, Y. Zhou, “A High Sensitivity Flexible Stretchable Fiber Optic Sensor based on the Waveguide Loss Principle,” *IEEE Transactions on Instrumentation and Measurement*, 2024.
- J. Liang, G. Lou, F. Zhou, **Yumeng Cai**, C. Wang, Y. Zhou, “Residual Reinforcement Learning Based on Inverse Kinematic Modeling for Soft Robotic Arm Control,” *ASME Journal of Dynamic Systems, Measurement, and Control*, 2024.
- H. Huang, S. Zhang, Y. Chen, S. Liu, Z. Chen, **Yumeng Cai**, H. Li, L. Xie, “Portable sEMG-Controlled Hand Electrical Stimulation System,” *IEEE Sensors Journal*, 2024.
- [†] Equal contribution.
- “A Gesture Recognition Smart Bracelet System Based on Surface Electromyography Signal,” Invention Patent, second inventor, 2023.
- “A Base Station Energy Storage Box,” Utility Model Patent, second inventor, 2023.

SELECTED RESEARCH EXPERIENCE

- Robotic Surgery Gap Discovery** May 2026 - Present
- Georgia Institute of Technology | Advisor: Prof. Jun Ueda
 - Developed an AI-assisted analysis pipeline to identify under-addressed functional needs and research gaps in robotic and laparoscopic surgery.
- Data-Driven Single-Cell Analysis of GRM5/Grm5-Associated Microglial States** Mar. 2026 - Present
- Georgia Institute of Technology | Advisor: Prof. Levi Wood
 - Analyzed mouse and human single-cell data to characterize GRM5/Grm5-associated microglial states.
 - Applied Spearman correlation and FDR correction to identify genes associated with Grm5 expression and compare canonical marker programs.
- Image-Guided Robotic Guidewire Navigation System** Sep. 2025 - Feb. 2026
- Georgia Institute of Technology | Advisor: Prof. Jaydev P. Desai
 - Designed and developed a bidirectionally steerable guidewire and its robotic control platform.
 - Developed a C-arm image-based perception pipeline for robust real-time guidewire tracking.
 - Developed an image-guided path-planning framework for guidewire navigation toward predefined targets.
- Miniature Hybrid-Actuated Soft Manipulator with FBG-Based Shape Sensing** Sep. 2024 - Mar. 2026

- *South China University of Technology* | *Advisor: Prof. Yitong Zhou*
- Designed a miniature soft continuum robot integrating cable-driven and pneumatic actuation for minimally invasive procedures.
- Developed an FBG-based shape-sensing pipeline using a data-driven prediction model and two-segment PCC modeling for shape reconstruction.
- Achieved independent 6-DoF motion in an 11-mm-diameter soft manipulator.

Soft Wearable Assistive Exosuit

Jul. 2024 - Sep. 2024

- *University of Notre Dame* | *Advisor: Prof. Margaret Coad*
- Investigated soft and bioinspired design, modeling, and control methods for wearable exosuits and prosthetic systems.
- Led system-level improvements, coordinated experimental evaluation with the health center, and collected data to improve comfort and usability.

Hydrodynamics of Quadrupedal Robotic Paddling

Sep. 2023 - Jul. 2024

- *South China University of Technology* | *Advisor: Prof. Ye Chen*
- Analyzed hydrodynamic forces, power consumption, and flow-field characteristics of single-leg and quadrupedal paddling across multiple gait patterns, including diagonal and lateral-sequence gaits, with varying swing angles, ranges, and phase relationships.
- Processed large-scale simulation datasets on Linux (Ubuntu) for quadrupedal paddling analysis.
- Post-processed and visualized CFD results using Tecplot.

Manta Ray-Inspired Soft Robotic Swimmer

Jun. 2023 - Sep. 2024

- *South China University of Technology* | *Advisor: Prof. Yitong Zhou*
- Designed a bistable soft swimmer using precurved composites and McKibben muscles for snap-through flapping.
- Evaluated swimming performance and demonstrated forward, reverse, turning, and flipping motions.

Stretchable Fiber-Optic Sensor for Large-Strain Sensing

Jan. 2023 - Jun. 2024

- *South China University of Technology* | *Advisor: Prof. Yitong Zhou*
- Fabricated a flexible, stretchable fiber optic sensor optimized for continuous health monitoring.
- Characterized sensing performance, including strain range, detection limit, and repeatability.
- Developed a human-machine interaction prototype that translated user motion into interaction signals using the sensor.

sEMG-Based Gesture Recognition Smart Bracelet

Jul. 2022 - May 2023

- *South China University of Technology* | *Advisor: Prof. Longhan Xie*
- Developed signal-processing and recognition methods to address weak wrist sEMG signals, noise interference, and inter-user variability in gesture recognition.
- Designed the PCB and array-based sEMG bracelet, deployed cloud-based training models, and developed a CNN-based gesture recognition network.

HONORS & AWARDS

Third Prize , Soft Robot Innovation Design Competition, 9th Soft Robotics Conference	Nov. 2024
Third Prize , National Undergraduate Electronics Design Contest	Aug. 2023
Third Prize , 22nd ROBOCON National College Student Robot Competition	Jul. 2023
Project Lead & Award Recipient , China International College Students' "Internet+" Innovation and Entrepreneurship Competition	Jun. 2023
Project Lead & Special Prize Recipient , Guangdong "Challenge Cup" College Students Extracurricular Academic Works Competition	May 2023

SKILLS & ADDITIONAL EXPERIENCE

Teaching	Teaching Assistant, Deformable Bodies (COE 3001), Georgia Tech, Summer 2026
Programming	Python, C/C++, MATLAB, R
Engineering Tools	SolidWorks, Altium Designer, Arduino, Tecplot, ANSYS Mechanical, Linux (Ubuntu)
Additional Experience	Machine Learning Winter Program, Girton College, University of Cambridge Supervised by Prof. Pietro Liò